

Exercises

3.14 Solve this system. Then solve the associated homogeneous system.

$$\begin{aligned}x + y - 2z &= 0 \\x - y &= -3 \\3x - y - 2z &= -6 \\2y - 2z &= 3\end{aligned}$$

✓ 3.15 Solve each system. Express the solution set using vectors. Identify a particular solution and the solution set of the homogeneous system.

$$\begin{array}{lll}(\text{a}) \quad 3x + 6y = 18 & (\text{b}) \quad x + y = 1 & (\text{c}) \quad x_1 + x_3 = 4 \\& x + 2y = 6 & x - y = -1 \quad x_1 - x_2 + 2x_3 = 5 \\& & 4x_1 - x_2 + 5x_3 = 17\end{array}$$

$$\begin{array}{lll}(\text{d}) \quad 2a + b - c = 2 & (\text{e}) \quad x + 2y - z = 3 & (\text{f}) \quad x + z + w = 4 \\2a + c = 3 & 2x + y + w = 4 & 2x + y - w = 2 \\a - b = 0 & x - y + z + w = 1 & 3x + y + z = 7\end{array}$$

3.16 Solve each system, giving the solution set in vector notation. Identify a particular solution and the solution of the homogeneous system.

$$\begin{array}{lll}(\text{a}) \quad 2x + y - z = 1 & (\text{b}) \quad x - z = 1 & (\text{c}) \quad x - y + z = 0 \\4x - y = 3 & y + 2z - w = 3 & y + w = 0 \\& x + 2y + 3z - w = 7 & 3x - 2y + 3z + w = 0 \\& & -y - w = 0\end{array}$$

$$\begin{array}{l}(\text{d}) \quad a + 2b + 3c + d - e = 1 \\3a - b + c + d + e = 3\end{array}$$

✓ 3.17 For the system

$$\begin{aligned}2x - y - w &= 3 \\y + z + 2w &= 2 \\x - 2y - z &= -1\end{aligned}$$

which of these can be used as the particular solution part of some general solution?

$$\begin{array}{lll}(\text{a}) \quad \begin{pmatrix} 0 \\ -3 \\ 5 \\ 0 \end{pmatrix} & (\text{b}) \quad \begin{pmatrix} 2 \\ 1 \\ 1 \\ 0 \end{pmatrix} & (\text{c}) \quad \begin{pmatrix} -1 \\ -4 \\ 8 \\ -1 \end{pmatrix}\end{array}$$

✓ 3.18 Lemma 3.7 says that we can use any particular solution for $\vec{p}$. Find, if possible, a general solution to this system

$$\begin{aligned}x - y + w &= 4 \\2x + 3y - z &= 0 \\y + z + w &= 4\end{aligned}$$

that uses the given vector as its particular solution.

$$\begin{array}{lll}(\text{a}) \quad \begin{pmatrix} 0 \\ 0 \\ 0 \\ 4 \end{pmatrix} & (\text{b}) \quad \begin{pmatrix} -5 \\ 1 \\ -7 \\ 10 \end{pmatrix} & (\text{c}) \quad \begin{pmatrix} 2 \\ -1 \\ 1 \\ 1 \end{pmatrix}\end{array}$$

3.19 One is nonsingular while the other is singular. Which is which?

$$\begin{array}{ll}(\text{a}) \quad \begin{pmatrix} 1 & 3 \\ 4 & -12 \end{pmatrix} & (\text{b}) \quad \begin{pmatrix} 1 & 3 \\ 4 & 12 \end{pmatrix}\end{array}$$

✓ 3.20 Singular or nonsingular?

$$(a) \begin{pmatrix} 1 & 2 \\ 1 & 3 \end{pmatrix} \quad (b) \begin{pmatrix} 1 & 2 \\ -3 & -6 \end{pmatrix} \quad (c) \begin{pmatrix} 1 & 2 & 1 \\ 1 & 3 & 1 \end{pmatrix} \quad (d) \begin{pmatrix} 1 & 2 & 1 \\ 1 & 1 & 3 \\ 3 & 4 & 7 \end{pmatrix}$$

$$(e) \begin{pmatrix} 2 & 2 & 1 \\ 1 & 0 & 5 \\ -1 & 1 & 4 \end{pmatrix}$$

✓ 3.21 Is the given vector in the set generated by the given set?

$$(a) \begin{pmatrix} 2 \\ 3 \end{pmatrix}, \left\{ \begin{pmatrix} 1 \\ 4 \end{pmatrix}, \begin{pmatrix} 1 \\ 5 \end{pmatrix} \right\}$$

$$(b) \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}, \left\{ \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \right\}$$

$$(c) \begin{pmatrix} 1 \\ 3 \\ 0 \end{pmatrix}, \left\{ \begin{pmatrix} 1 \\ 0 \\ 4 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \\ 5 \end{pmatrix}, \begin{pmatrix} 3 \\ 3 \\ 0 \end{pmatrix}, \begin{pmatrix} 4 \\ 2 \\ 1 \end{pmatrix} \right\}$$

$$(d) \begin{pmatrix} 1 \\ 0 \\ 1 \\ 1 \end{pmatrix}, \left\{ \begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 3 \\ 0 \\ 0 \\ 2 \end{pmatrix} \right\}$$

3.22 Prove that any linear system with a nonsingular matrix of coefficients has a solution, and that the solution is unique.

3.23 In the proof of Lemma 3.6, what happens if there are no non- $0 = 0$ equations?

✓ 3.24 Prove that if $\vec{s}$ and $\vec{t}$ satisfy a homogeneous system then so do these vectors.

$$(a) \vec{s} + \vec{t} \quad (b) 3\vec{s} \quad (c) k\vec{s} + m\vec{t} \text{ for } k, m \in \mathbb{R}$$

What's wrong with this argument: "These three show that if a homogeneous system has one solution then it has many solutions—any multiple of a solution is another solution, and any sum of solutions is a solution also—so there are no homogeneous systems with exactly one solution."

3.25 Prove that if a system with only rational coefficients and constants has a solution then it has at least one all-rational solution. Must it have infinitely many?